Hari haran Suresh kumar

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Professional Summary

Data Science graduate with hands-on experience in machine learning, data engineering, and analytics across healthcare and financial domains. Skilled in building end-to-end ML pipelines, optimizing SQL systems, and developing scalable data workflows. Proven ability to translate complex data into actionable insights through collaboration with cross-functional stakeholders.

Education

University of Washington — Seattle, WA
M.S. in Data Science | GPA: 3.86 — *September 2024 – March 2026*

Professional Experience

Virginia Mason Franciscan Health — Seattle, WA
Data Science Intern — *September 2025 – March 2026*

- Built an end-to-end ML pipeline on 275 patient records using Ridge Regression, XGBoost, and Random Forest to predict postoperative shifts across 7 spinal alignment parameters, replacing a manual process where surgeons individually derived spinopelvic measurements from patient X-rays.
- Engineered a genetic algorithm optimization framework (PyMoo) evaluating 1,000+ candidate surgical plans per patient across 10 generations, producing ranked outputs across 9 clinician-defined scenarios to enable systematic comparison of alignment outcomes and mechanical failure trade-offs.
- Designed a composite scoring function integrating 7 clinical components including GAP score, 4 biomechanical constraint penalties, and mechanical failure probability, providing a structured objective basis for surgical plan comparison where no such framework previously existed.
- Collaborated with 9 surgeons to validate model outputs using SHAP plots, predicted vs. actual scatter plots, and confusion matrices, iterating on feature engineering and constraint formulation based on expert clinical feedback.

Cognizant — Chennai, India
*Analyst – **US Bank*** — *July 2022 – March 2024*

- Optimized SQL queries within the KYT internal analytics tool supporting US Bank, reducing processing time by 30% for daily execution and enabling 25+ downstream teams to analyze hierarchy, velocity, and workflow metrics through query rewriting and indexing.
- Partnered with 3 data engineers to audit 3 large-scale relational schemas (20+ tables each with foreign key dependencies), removing 4 inactive standalone tables per schema and producing a structured data dictionary and ER documentation serving 25+ downstream US Bank teams.

*Analyst Trainee – **US Bank*** — *January 2022 – June 2022*

- Managed biweekly data ingestion workflows processing 50+ new employee records per week into PostgreSQL, ensuring data availability for internal analytics tools using Excel-based preprocessing and database loading.
- Monitored production systems and resolved UI and data inconsistencies through JIRA ticketing, reducing data-related issues through proactive identification and debugging.

Intern — *August 2021 – December 2021*

- Built a full-stack claims management system prototype with user and claims tracking modules, enabling visibility into accepted and denied claims through React.js and PostgreSQL-based system design.

Projects

AWS Event Analytics Pipeline | *AWS (S3, Lambda, Glue, Athena), PySpark, Parquet* | Report — September 2025

- Built a serverless Bronze-to-Silver ETL pipeline ingesting 500K–750K e-commerce events per batch via Lambda and EventBridge, using Glue job bookmarks for incremental processing without reloading prior data, and partitioning output as Parquet to reduce Athena scan costs across five business analytics queries covering funnels, revenue, and user engagement.

Seattle Traffic Accident Dashboard | Dashboard — September 2024

- Designed an interactive Tableau dashboard analyzing Seattle traffic accident data, enabling identification of seasonal trends and spatial hotspots through geospatial visualization; presented to MS Data Science cohort.

Hospital Billing Variation Analysis | GitHub — September 2025

- Analyzed NY State inpatient discharge data to study hospital billing variation, identifying insurance payer type as a major driver of cost differences across similar cases using regression and ANOVA.

Technical Skills

Programming & Data Processing: Python, SQL (PostgreSQL), Pandas, PySpark, Databricks
Data Engineering: ETL Pipelines, Data Modeling, Data Validation, Data Quality Checks, AWS (S3, Lambda, Glue, Athena), Apache Spark, Parquet, Snowflake
Analytics & Machine Learning: Regression (Linear & Logistic), Tree-based Models (XGBoost, Random Forest), Feature Engineering, A/B Testing, Causal Inference, PyMoo, Scikit-learn
Visualization & Tools: Tableau, Power BI, Matplotlib, Seaborn, Git, JIRA, Azure